Credit Decision Data Architecture & Implementation

Technical Proposal for MAL AI-Native Digital Bank

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Executive Summary

As MAL expands its digital lending portfolio through products such as Personal Finance, Buy Now Pay Later (BNPL), and Credit Card Alternatives, the ability to make fast, reliable, and compliant credit decisions becomes a key business capability. Each lending decision depends on data collected from multiple internal and external systems, where information is received in different formats, at different times, and with varying levels of quality.

The proposed solution is a cloud-native data architecture that consolidates these diverse data sources into a unified and trusted platform for credit decisioning. The design follows the Medallion Architecture (Bronze, Silver, and Gold) to ensure data is ingested securely, transformed consistently, and delivered in a business-ready format while preserving complete traceability of every lending decision.

The architecture prioritizes scalability, data quality, auditability, and operational resilience. It incorporates AWS services such as Amazon S3, AWS Glue, Apache Airflow, and PostgreSQL to support both batch and event-driven data processing. Automated validation, customer identity resolution, and comprehensive monitoring are integrated throughout the pipeline to improve data reliability and support regulatory compliance.

This document presents the proposed architecture, explains key design decisions, outlines data quality and governance mechanisms, and demonstrates how the platform can support current business needs while remaining scalable for future growth.

Business Problem & Solution Overview

2. Business Context

MAL AI-Native Digital Bank is introducing a new generation of digital lending products, including Personal Finance, Buy Now Pay Later (BNPL), and Credit Card Alternatives. To make accurate and timely credit decisions, the bank must combine information from multiple internal and external systems, each providing a different perspective on the applicant's financial profile and risk.

These systems deliver data in different formats (XML, JSON, webhooks, and relational databases), use different customer identifiers, and operate at different speeds. Without a unified data platform, the bank may face inconsistent customer records, delayed decisions, increased operational complexity, and challenges in meeting regulatory and audit requirements.

2.1 Key Challenges

The proposed solution addresses the following challenges:

- Integrating data from multiple heterogeneous sources.
- Standardizing different data formats into a common data model.
- Resolving customer identities across systems using different identifiers.
- Preserving historical records for auditability and regulatory compliance.
- Ensuring high data quality before data is consumed by downstream applications.
- Supporting a scalable architecture capable of handling increasing application volumes.

2.2 Proposed Solution

To address these challenges, I propose a cloud-native data engineering architecture built on AWS using the Medallion Architecture (Bronze, Silver, and Gold).

The solution ingests raw data from all source systems into a centralized data lake, applies validation and transformation processes to produce standardized datasets, resolves customer identities through defined matching rules, and delivers trusted, business-ready data for credit decisioning and portfolio analytics.

This architecture is designed with scalability, security, and governance in mind, ensuring that every credit decision is traceable, reproducible, and supported by high-quality data.

Source Systems

Source System	Data Format	Ingestion Method	Primary Identifier
UAE Credit Bureau	XML	SFTP	Emirates ID
Fraud Detection Provider	JSON	REST API	Phone + Email
AML / PEP Screening	JSON	Webhook	Name + Date of Birth
Internal Customer Profile	PostgreSQL	Database Connection	Customer UUID

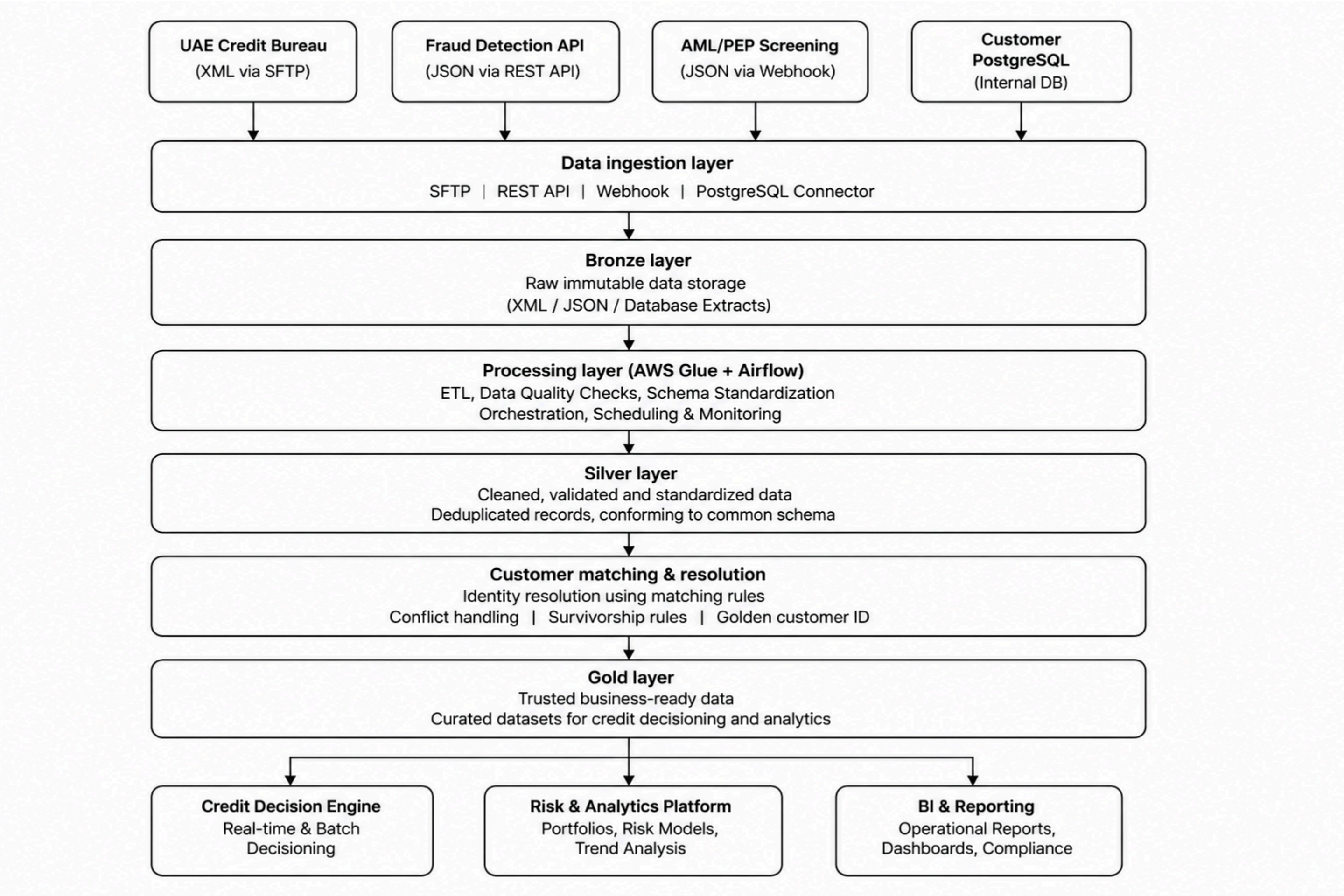
Proposed Data Architecture

Proposed Solution Architecture

Overview

The proposed solution adopts a cloud-native architecture on AWS to ingest, process, and consolidate data from multiple internal and external sources into a unified platform for credit decisioning. The architecture supports both batch and event-driven data ingestion while ensuring scalability, data quality, traceability, and regulatory compliance.

Data is first collected from all source systems and stored in the Bronze layer without modification. It then progresses through transformation and validation processes in the Silver layer before being published as trusted business datasets in the Gold layer. The final datasets are consumed by the Credit Decision Engine, Risk Analytics, and operational reporting systems.



Key Components

Component	Purpose
Data Sources	Provide customer, credit, fraud, and AML information.
Data Ingestion Layer	Collects data from SFTP, REST APIs, webhooks, and PostgreSQL.
Amazon S3 Bronze	Stores raw data exactly as received for auditing and replay.
AWS Glue	Performs data transformation, cleansing, and schema standardization.
Apache Airflow	Orchestrates ETL workflows, scheduling, retries, and dependency management.
Amazon S3 Silver	Stores validated and standardized datasets.
Customer Matching Engine	Resolves customer identities and handles conflicting records.
Amazon S3 Gold	Provides curated datasets ready for credit decisioning and analytics.
Business Consumers	Credit Decision Engine, Risk Team, and Business Intelligence dashboards.

Architecture Highlights :

The proposed architecture has been designed to achieve the following objectives:

- Integrate multiple heterogeneous data sources into a unified platform.
- Preserve raw source data to maintain complete auditability.
- Improve data consistency through standardized transformation pipelines.
- Enable reliable customer identity resolution across systems.
- Deliver trusted datasets for lending decisions and portfolio monitoring.
- Support future business growth through a scalable cloud-native design.

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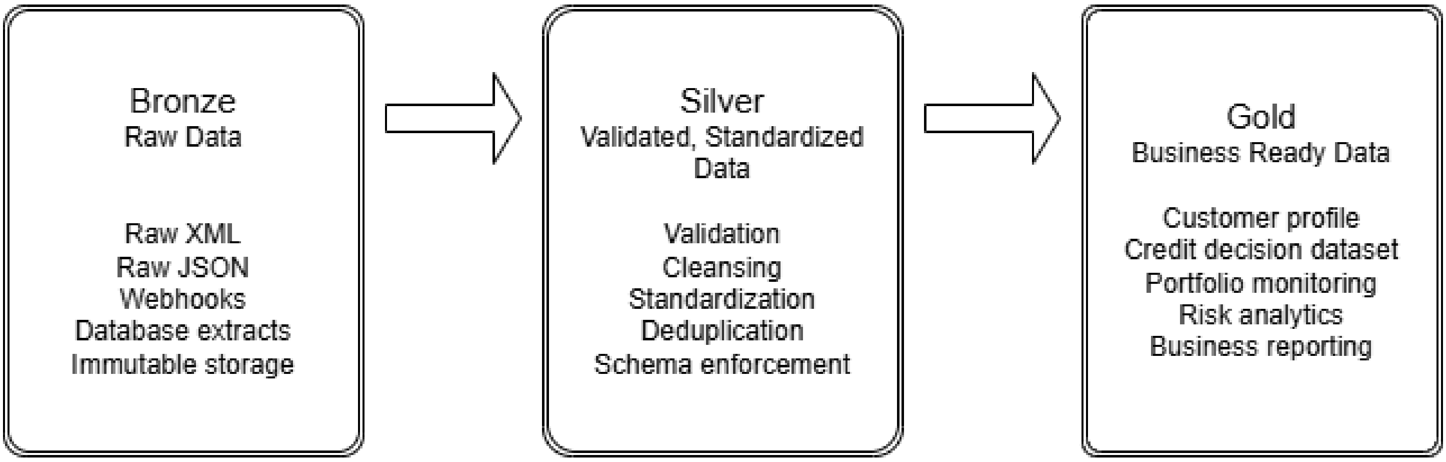
4. Medallion Data Architecture

Overview

To establish a reliable and scalable foundation for credit decisioning, the proposed solution follows the Medallion Architecture pattern, consisting of Bronze, Silver, and Gold layers. This layered approach progressively improves data quality while preserving the original source data for auditability and regulatory compliance.

Each layer has a clearly defined responsibility, enabling efficient data processing, easier troubleshooting, and consistent delivery of trusted datasets to downstream consumers.

Medallion Data Flow Diagram



Layer Responsibilities

Layer	Purpose	Example Output
Bronze	Stores raw source data exactly as received without transformation	Raw XML bureau files, Fraud API responses, AML webhooks
Silver	Applies validation, cleansing, schema standardization, and duplicate removal.	Standardized customer records with validated attributes
Gold	Produces business-ready datasets for credit decisioning, analytics, and dashboards.	Unified customer profile, credit decision dataset, portfolio metrics

This section can be visualized as a timeline or bar chart in Canva.

Data Processing Strategy

The processing pipeline follows a structured sequence to maintain consistency across all incoming data sources.

1. **Ingestion** – Raw data is collected from SFTP, REST APIs, webhooks, and PostgreSQL.
2. **Bronze Layer** – All source data is stored in its original format with ingestion timestamps and metadata.
3. **Silver Layer** – Data undergoes quality validation, schema normalization, duplicate detection, and enrichment before being standardized into a common structure.
4. **Gold Layer** – Trusted datasets are generated to support lending decisions, portfolio monitoring, and business intelligence.

This layered design ensures that downstream applications consume only validated and standardized data while preserving complete traceability back to the original source records.

Benefits of the Medallion Architecture

- Preserves original source data for audit and regulatory compliance.
- Isolates raw, processed, and business-ready datasets.
- Improves data quality before downstream consumption.
- Simplifies troubleshooting and data lineage.
- Supports scalable processing as application volumes increase.
- Enables reliable reporting and credit decision consistency

Customer Identity Resolution & Decision Traceability

Customer Identity Resolution

Overview

Customer information is received from multiple systems, each using different identifiers. To create a unified customer profile, the platform applies a deterministic matching strategy that prioritizes the most reliable identifiers before considering secondary attributes.

The matching process follows a predefined hierarchy to minimize duplicate records and ensure consistent customer identification across all lending products.

Priority	Identifier	Source
1	Internal Customer UUID	PostgreSQL
2	Emirates ID	UAE Credit Bureau
3	Phone + Email	Fraud Detection API
4	Name + Email ID	AML / PEP Screening
5	Manual Review	Unmatched Records

New Customer Record

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▼

Internal UUID Available?

│ │

Yes No

│

Match

│

Complete

│

No

▼

Emirates ID Match?

│

Yes → Match

│

No

▼

Phone + Email Match?

|

Yes → Match

|

No

▼

Name + DOB Match?

|

Yes → Match

|

No

▼

Manual Review Queue

5.2 Decision Traceability

Every credit decision must be reproducible for regulatory audits. The platform therefore captures an immutable snapshot of all decision inputs, including source data, transformations, matching results, and decision metadata.

Each decision record includes:

- Decision ID
- Customer ID
- Input datasets
- Source timestamps
- Credit bureau snapshot
- Fraud score
- AML screening result
- Processing timestamp
- Model/version used
- Final decision outcome

This approach enables complete auditability, supports investigations, and satisfies UAE Central Bank compliance requirements.

Data Quality Framework & Portfolio Monitoring Mart

6. Data Quality Framework

Overview

Accurate credit decisions depend on high-quality, trustworthy data. To ensure reliability across all lending products, the proposed platform implements automated data quality checks at the Silver layer before records are promoted to the Gold layer. Critical validation rules are treated as mandatory ("must-pass"), while operational metrics are continuously monitored using configurable alert thresholds.

Any record failing a must-pass validation is quarantined for investigation and excluded from downstream credit decision processing until the issue is resolved.

Data Quality Scorecard

Validation Rule	Description	Severity	Alert Threshold	Action
Emirates ID must not be null	Mandatory customer identifier	Critical	> 0 failed records	Reject record
Customer UUID uniqueness	Prevent duplicate customer profiles	Critical	> 1 duplicate	Quarantine
Fraud score availability	Fraud API response received	Critical	> 5% missing	Alert & Retry
AML screening status	AML response must be available	Critical	> 5% missing	Hold decision
XML/JSON schema validation	Validate incoming data structure	High	> 2% failures	Alert operations
Data freshness	Bureau data not older than 24 hours	Medium	> 24 hours	Warning & reload

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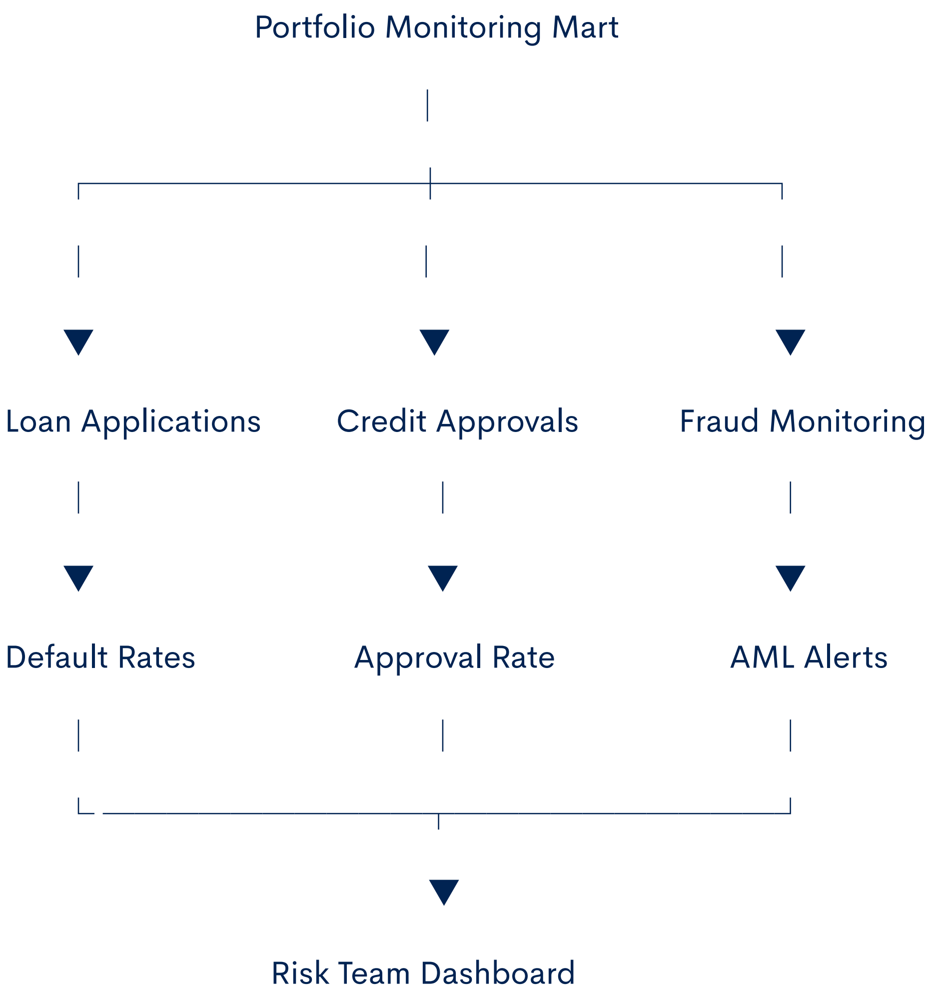
Portfolio Monitoring Mart

Overview

The Gold layer publishes curated datasets into a Portfolio Monitoring Mart that provides the Risk and Business teams with a consolidated view of lending performance. This mart supports operational monitoring, portfolio analysis, and executive reporting without impacting transactional systems.

Portfolio Monitoring Dashboard



Key Business Metrics

Category	Example Metrics
Lending	Applications Received, Approval Rate, Rejection Rate
Customer	New Customers, Repeat Customers
Risk	Default Rate, Average Credit Score
Fraud	Fraud Alerts, Fraud Detection Rate
AML	Pending Reviews, Positive Matches
Operations	Data Processing Time, ETL Success Rate

Benefits

- Detects data issues before they impact lending decisions.
- Provides real-time visibility into portfolio performance.
- Supports proactive monitoring of fraud and AML activities.
- Enables faster operational response through automated alerts.
- Delivers trusted business metrics for management reporting.
- Improves governance and regulatory compliance.

Production Readiness, Scalability & Security

7. Production Readiness

Overview

The proposed architecture is designed not only to meet the initial launch requirements of approximately **10,000 credit decisions per day** but also to scale efficiently to **100,000 daily decisions within the next 12 months**. The platform leverages managed AWS services, modular data pipelines, and automated monitoring to ensure high availability, operational resilience, and maintainability while complying with UAE Central Bank regulatory expectations.

Scalability Strategy

Area	Design Approach	Business Benefit
Data Storage	Amazon S3 Data Lake with partitioned datasets	Cost-effective and virtually unlimited storage
Data Processing	AWS Glue ETL with parallel processing	Faster execution for increasing workloads
Workflow Orchestration	Apache Airflow with retry and dependency management	Reliable pipeline execution
Database	Amazon RDS PostgreSQL	High availability for operational metadata
Monitoring	Amazon CloudWatch alerts and dashboards	Early detection of failures and performance issues
Future Growth	Modular architecture supporting additional lending products	Simplifies expansion without major redesign

Security & Compliance Controls

Control	Implementation
Encryption at Rest	Amazon S3 and RDS encryption using AWS KMS
Encryption in Transit	HTTPS, TLS for APIs and SFTP transfers
Identity & Access Management	AWS IAM roles following least privilege principle
Audit Trail	Immutable storage of decision inputs and processing logs
Data Lineage	End-to-end tracking from raw ingestion to Gold datasets
Regulatory Compliance	Complete decision traceability supporting UAE Central Bank audit requirements

Operational Monitoring

The platform continuously monitors operational health and data pipeline performance through automated alerts and dashboards.

Key operational metrics include:

- ETL job success and failure rates
- API response latency
- Data ingestion volume
- Processing duration
- Data quality validation failures
- Infrastructure resource utilization
- Daily credit decisions processed

Risk Mitigation Strategy

Risk	Mitigation
Source system unavailable	Automatic retry mechanism and alert notifications
Corrupted input files	Schema validation before processing
Duplicate customer records	Multi-level customer matching logic
ETL job failure	Airflow retry policy with failure alerts
Data quality degradation	Automated quality scorecards and quarantine process

Key Design Principles

- **Scalability** – Supports increasing transaction volumes without major architectural changes.
- **Reliability** – Automated retries and monitoring reduce operational risk.
- **Security** – Strong encryption and controlled access protect sensitive customer data.
- **Auditability** – Every credit decision is fully traceable from source to outcome.
- **Maintainability** – Modular ETL pipelines simplify future enhancements and operational support.

Conclusion & Recommendations

8. Conclusion

The proposed architecture provides a scalable, secure, and cloud-native foundation for MAL's credit decision platform. By integrating data from multiple internal and external sources into a unified Medallion Architecture, the solution delivers trusted, high-quality datasets that support faster and more reliable lending decisions.

The design emphasizes data quality, customer identity resolution, decision traceability, and operational resilience while ensuring that every credit decision remains fully auditable. The use of managed AWS services simplifies operations, enables future scalability, and supports the bank's objective of delivering AI-driven financial products efficiently and securely.

This architecture is designed not only to address the current lending requirements but also to provide a flexible platform that can accommodate future products, increased transaction volumes, and evolving regulatory expectations.

Key Design Decisions

Design Decision	Business Value
Medallion Architecture	Separates raw, validated, and business-ready data for better governance and maintainability.
Amazon S3 Data Lake	Provides scalable and cost-effective storage for structured and semi-structured data.
AWS Glue & Apache Airflow	Enables reliable ETL processing, orchestration, and workflow automation.
Customer Identity Resolution	Reduces duplicate records and improves consistency across systems.
Data Quality Framework	Ensures only validated data is used for credit decisioning.
Immutable Decision Records	Supports auditability and regulatory compliance.

Future Enhancements

The proposed platform can be extended with additional capabilities as business requirements evolve.

- Real-time streaming data ingestion using Amazon Kinesis or Apache Kafka.
- Machine learning models for fraud detection and credit risk prediction.
- Self-service analytics through Amazon QuickSight dashboards.
- Automated metadata management and data cataloging using AWS Glue Data Catalog.
- Enhanced observability with centralized logging and performance dashboards.

Final Remarks

This solution has been designed with a strong focus on scalability, reliability, security, and compliance while keeping the architecture modular and easy to maintain. By combining standardized data engineering practices with AWS cloud services, the platform establishes a reliable foundation for intelligent credit decisioning and future innovation within MAL AI-Native Digital Bank.

Acknowledgement

Thank you for reviewing this technical proposal. I appreciate the opportunity to present my approach and look forward to discussing the design decisions and implementation strategy in greater detail.